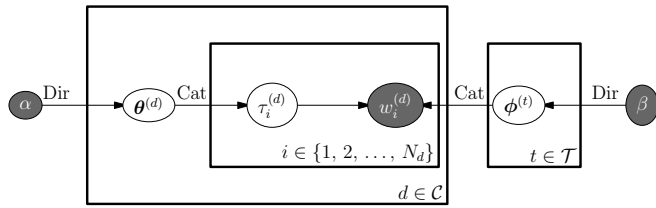


Graphical Models



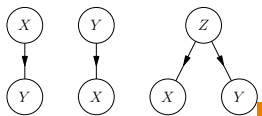
Conditional Independence, Graphical models, LDA

Graphical Models

- If we want to build large probabilistic inference systems
 - ★ AI Doctor
 - ★ Fault diagnostic system for a computer
- we can describe this by introducing random variables, but it is helpful to graphically represent causal connections
- Graphical models allow us to do this
 - It allows us to build a joint probability distribution from which we can compute everything we want

Graphical Models

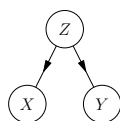
- **Bayesian Belief Networks** are a type of graphical models where we use a directed graphs to show causal relationships between random variables
 - We could represent the three conditions described above by
-
- ```
graph TD; X1((X)) --> Y1((Y)); Y2((Y)) --> X2((X)); Z((Z)) --> X3((X)); Z --> Y3((Y));
```
- We can use these graphical representations to work out how to efficiently average over latent variables



## Conditional Independence

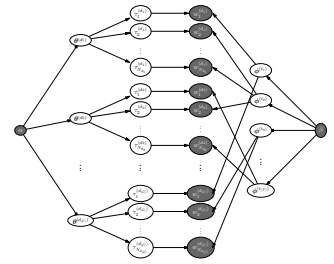
- A weaker notion is conditional independence

$$\mathbb{P}(X,Y|Z) = \mathbb{P}(X|Z)\mathbb{P}(Y|Z) \blacksquare$$



- Conditional independence implies that there is no direct causation
- But it doesn't imply zero correlation
- Conditional independence reduces computational complexity, e.g.

$$\mathbb{E}[XY] = \sum_{X,Y,Z} XY \mathbb{P}(X,Y,Z) = \sum_Z P(Z) \left( \sum_X XP(X|Z) \right) \left( \sum_Y YP(Y|Z) \right)$$



1. **Graphical Models**
2. Cakes!
3. Latent Dirichlet Allocation

## Dependencies Between Variables

- In building a probabilistic model we want to know which random variables depend on each other directly and which don't
- Variables that don't will typically still be correlated
- If two random variables  $X$  and  $Y$  are correlated then
  - ★  $X$  could affect  $Y$
  - ★  $Y$  could affect  $X$
  - ★  $X$  and  $Y$  could not influence each other, but both be affected by another random variable  $Z$

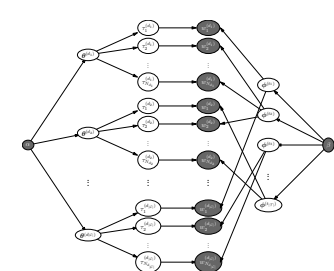
## Statistical Independence

- Two random variables are statistically independent if

$$\mathbb{P}(X,Y) = \mathbb{P}(X)\mathbb{P}(Y) \blacksquare$$

- Equally this implies  $\mathbb{P}(X|Y) = \mathbb{P}(X)$  and  $\mathbb{P}(Y|X) = \mathbb{P}(Y)$  ■
- Statistically independent variables are uncorrelated ■
- But statistical independence is often too powerful ■

## Outline



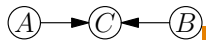
1. Graphical Models
2. **Cakes!**
3. Latent Dirichlet Allocation

## Let Them Eat Cakes

- I will go through a very simple example involving cakes
- It illustrates some simple principles
- In the subsidiary notes I present a very simple program for computing all the probabilities—I would encourage you to test this out, as it makes things much clearer

### Computing with Probabilities

- Other probabilities I can deduce, e.g.  
 $\mathbb{P}(C = 0|A, B) = 1 - \mathbb{P}(C = 1|A, B)$
- I can depict the causal relationship as



- The quantity that I really want is the joint probability  

$$\begin{aligned} \mathbb{P}(A, B, C) &= \mathbb{P}(C, B|A) \mathbb{P}(A) \\ &= \mathbb{P}(C|A, B) \mathbb{P}(B|A) \mathbb{P}(A) = \mathbb{P}(C|A, B) \mathbb{P}(B) \mathbb{P}(A) \end{aligned}$$
- Because  $\mathbb{P}(B|A) = \mathbb{P}(B)$

### Are There Any Cakes Left?

- We can use our model to compute the probabilities of there being cakes in the coffee room

$$\begin{aligned} \mathbb{P}(C = 1) &= \sum_{A, B, C \in \{0, 1\}} \mathbb{I}[C = 1] \mathbb{P}(A, B, C) \\ &= \sum_{A, B \in \{0, 1\}} \mathbb{P}(C = 1|A, B) \mathbb{P}(A) \mathbb{P}(B) = 0.29 \end{aligned}$$

- The probability that Abi baked a cake is just 0.2 and for Ben its 0.1 (which is what we assume at the start)
- The probability of them both baking on a particular day is 0.02

### Who Made Those Cakes?

- If we observe there are cakes

$$\mathbb{P}(A, B|C = 1) = \mathbb{P}(A, B, C = 1) / \mathbb{P}(C = 1)$$

- A straightforward if tedious calculation shows

$$\begin{aligned} \mathbb{P}(A = 1|C = 1) &= 0.628, \quad \mathbb{P}(B = 1|C = 1) = 0.317 \\ \mathbb{P}(A = 1, B = 1|C = 1) &= 0.069 \end{aligned}$$

- Note  $\mathbb{P}(A = 1, B = 1|C = 1) \neq \mathbb{P}(A = 1|C = 1) \mathbb{P}(B = 1|C = 1)$
- When we observe  $C$  then  $A$  and  $B$  are no longer independent

## The Cake Scenario

- Abi and Ben both bake cakes and bring them into the coffee room
- Abi will bring in cakes 20% of the time:  $\mathbb{P}(A = 1) = 0.2$
- Ben will bring in cakes 10% of the time:  $\mathbb{P}(B = 1) = 0.1$
- 90% of the time if either Abi or Ben have put cakes in the coffee room there is some left when I enter  
 $\mathbb{P}(C = 1|A = 1, B = 0) = \mathbb{P}(C = 1|A = 0, B = 1) = 0.9$
- If they both make cake then there is always cake left  
 $\mathbb{P}(C = 1|A = 1, B = 1) = 1$
- If neither Abi or Ben has made cake there is still a 5% chance someone else has put cake in the coffee room  
 $\mathbb{P}(C = 1|A = 0, B = 0) = 0.05$

### Computing Expectations

- By using the joint probability and summing over all unknown quantities, we can compute expectations of anything we are interested in
- These sums are often sped up using knowledge of conditional independence
- To compute the probability of an event  $\mathcal{E}$  we introduce an indicator function  $\mathbb{I}[\mathcal{E}]$  which is equal to 1 if the event happens and 0 otherwise  

$$\mathbb{P}(\mathcal{E}) = \mathbb{E}[\mathbb{I}[\mathcal{E}]]$$
- If  $E$  is a random variable equal to 1 if event  $\mathcal{E}$  happens and 0 otherwise then  $E = \mathbb{I}[\mathcal{E}]$

### Making Observation

- Making observations changes probabilities
- In graphical models observed random variables are shaded



- The probabilities conditioned on  $C$  is given by

$$\mathbb{P}(A, B|C) = \frac{\mathbb{P}(A, B, C)}{\mathbb{P}(C)}$$

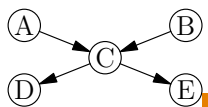
where

$$\mathbb{P}(C) = \sum_{A, B \in \{0, 1\}} \mathbb{P}(A, B, C)$$

### Elaborate Cakes

- We can elaborate on our cake model
- We suppose that Dave likes cakes so if there is a cake in the coffee room there is a 80% chance that I will see him eating a cake:  $\mathbb{P}(D = 1|C = 1) = 0.8$
- Even if there are no cakes in the coffee room there is a 10% chance that Dave has bought his own cake:  
 $\mathbb{P}(D = 1|C = 0) = 0.1$
- Eli also likes cakes: there is a 60% chance that I will see her eating cakes if there are cakes in the coffee room:  
 $\mathbb{P}(E = 1|C = 1) = 0.6$
- But she never buys herself cakes  $\mathbb{P}(E = 1|C = 0) = 0$

- We can depict this situation as



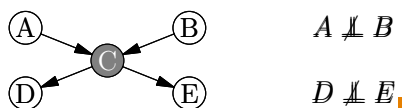
- This allows us to break down the joint probability

$$\begin{aligned}\mathbb{P}(A, B, C, D, E) &= \mathbb{P}(C, D, E | A, B) \mathbb{P}(B) \mathbb{P}(A) \\ &= \mathbb{P}(D | C) \mathbb{P}(E | C) \mathbb{P}(C | A, B) \mathbb{P}(B) \mathbb{P}(A)\end{aligned}$$

- We use the conditional independence of  $D$  and  $E$  given  $C$

## Observations and Independence

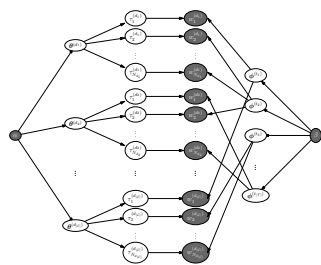
- Making observations changes the probabilities and in some case the dependencies of random variables on each other



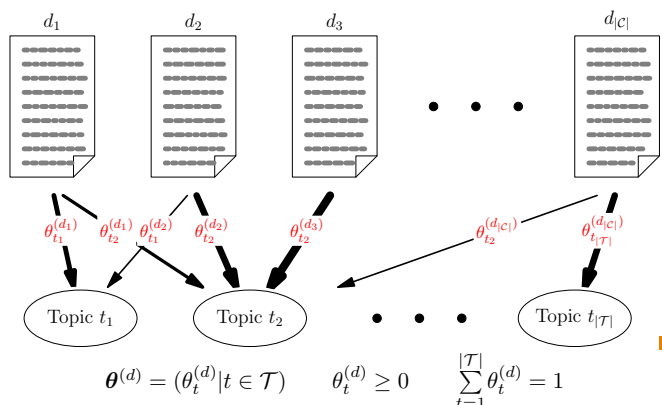
- There are rules to deduce the conditional independence from a graphical model given which variables have been observed—but these are details that you can look up if needed

## Outline

- Graphical Models
- Cakes!
- Latent Dirichlet Allocation



## Documents and Topic



- If we don't observe cakes then the probability of Dave and Eli eating cake are not independent

$$\begin{aligned}\mathbb{P}(D = 1) &= 0.303, & \mathbb{P}(E = 1) &= 0.174 \\ \mathbb{P}(D = 1, E = 1) &= 0.1392\end{aligned}$$

$$\text{so } \mathbb{P}(D, E) \neq \mathbb{P}(D) \mathbb{P}(E)$$

- This changes if we know there are cakes in the coffee room

$$\begin{aligned}\mathbb{P}(D = 1 | C = 1) &= 0.8 & \mathbb{P}(E = 1 | C = 1) &= 0.6 \\ \mathbb{P}(D = 1, E = 1 | C = 1) &= 0.48\end{aligned}$$

$$\text{so } \mathbb{P}(D = 1, E = 1 | C = 1) = \mathbb{P}(D = 1 | C = 1) \mathbb{P}(E = 1 | C = 1)$$

## Graphical Model Frameworks

- There are sophisticated frameworks for computing probabilities in Bayesian Belief Networks efficiently
- If our graph is a tree then we can evaluate probabilities efficiently
- When there are loops (so that a random variable both influences and is influenced by another random variables) then exact evaluation of expectations requires exhaustive summing over variables (which is often not tractable)
- There are various message passing algorithms designed to obtain approximations of expectations

## Model for Documents

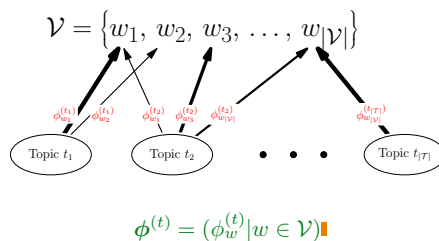
- We consider a model for the words in a set of documents (we ignore word order)
- We consider a corpus  $\mathcal{C} = \{d_i | i = 1, 2, \dots, |\mathcal{C}|\}$
- With documents consisting of words

$$d = (w_1^{(d)}, w_2^{(d)}, \dots, w_{N_d}^{(d)})$$

- We assume that there is a set of topics  $\mathcal{T} = \{t_1, t_2, \dots, t_{|\mathcal{T}|}\}$
- We associate a probability,  $\theta_t^{(d)}$ , that a word in document  $d$  relates to a topic  $t$

## Words and Topic

- We associate a probability  $\phi_w^{(t)}$  that a word,  $w$ , is related to a topic  $t$



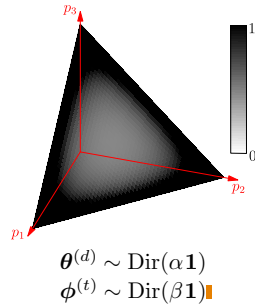
## Dirichlet Allocation

- Most documents are predominantly about a few topics and most topic have a small number of words associated to them

- We can generate sparse vectors  $\theta^{(d)}$  and  $\phi^{(t)}$  from a Dirichlet distribution with small parameters  $\alpha$

$$\text{Dir}(p|\alpha) = \Gamma\left(\sum_i \alpha_i\right) \prod_{i=1}^n \frac{p_i^{\alpha_i-1}}{\Gamma(\alpha_i)}$$

- $\sum_i p_i = 1$



## Generating Document

- To generate a document we choose a topic for each word and a word for each topic

$$\forall d \in \mathcal{C} \quad \theta^{(d)} \sim \text{Dir}(\alpha \mathbf{1})$$

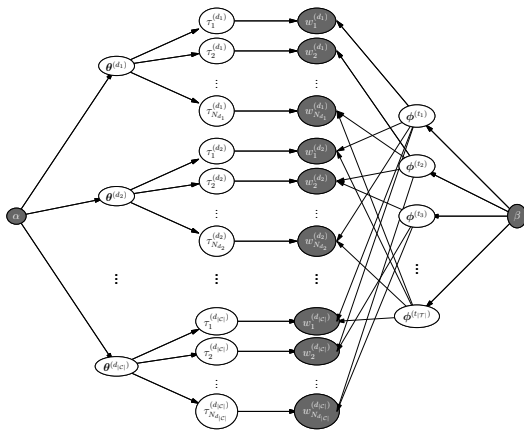
$$\forall t \in \mathcal{T} \quad \phi^{(t)} \sim \text{Dir}(\beta \mathbf{1})$$

$$\forall d \in \mathcal{C} \wedge \forall i \in \{1, 2, \dots, N_d\} \quad \tau_i^{(d)} \sim \text{Cat}(\theta^{(d)}), w_i^{(d)} \sim \text{Cat}(\phi^{(\tau_i^{(d)})})$$

- Where  $\text{Cat}(i|\mathbf{p}) = p_i$  is the categorical distribution (we choose one of a number of options)

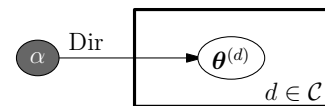
- This model is known as **Latent Dirichlet Allocation**

## LDA Graphical Model (version 1)



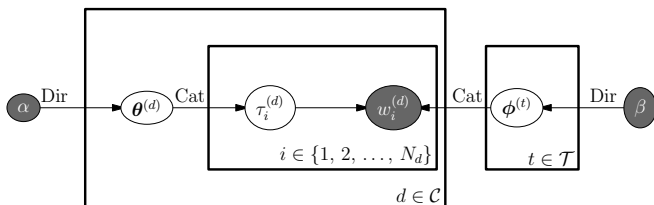
## Plate Diagrams

- Drawing every random variable is tedious (and not really possible)
- A short-hand is to draw a box (plate) meaning repeat



- That is we generate vectors  $\theta^d$  from a Dirichlet distribution  $\text{Dir}(\theta|\alpha \mathbf{1})$  for all documents in corpus  $\mathcal{C}$

## LDA Graphical Model (version 2)



- This is a lot more compact
- Personally, I find it hard to read, but you get used to it

## Probabilistic Model

- The graphical Model is shorthand for the variables

$$\mathbf{W} = (w_i^{(d)} | d \in \mathcal{C}) \quad \text{with} \quad w_i^{(d)} = (w_1^{(d)}, w_2^{(d)}, \dots, w_{N_d}^{(d)}), \quad \text{and} \quad w_i^{(d)} \in \mathcal{V}$$

$$\mathbf{T} = (\tau_i^{(d)} | d \in \mathcal{C} \wedge i \in \{1, 2, \dots, N_d\}) \quad \text{with} \quad \tau_i^{(d)} \in \mathcal{T}$$

$$\Theta = (\theta^{(d)} | d \in \mathcal{C}) \quad \text{with} \quad \theta^{(d)} = (\theta_t^{(d)} | t \in \mathcal{T}) \in \Lambda^{|\mathcal{T}|}$$

$$\Phi = (\phi^{(t)} | t \in \mathcal{T}) \quad \text{with} \quad \phi^{(t)} = (\phi_w^{(t)} | w \in \mathcal{V}) \in \Lambda^{|\mathcal{V}|}$$

- Distributed according to

$$\mathbb{P}(\mathbf{W}, \mathbf{T}, \Theta, \Phi | \alpha, \beta) = \left( \prod_{t \in \mathcal{T}} \text{Dir}(\phi^{(t)} | \beta \mathbf{1}) \right) \left( \prod_{d \in \mathcal{C}} \text{Dir}(\theta^{(d)} | \alpha \mathbf{1}) \prod_{i=1}^{N_d} \text{Cat}(\tau_i^{(d)} | \theta^{(d)}) \text{Cat}(w_i^{(d)} | \phi^{(\tau_i^{(d)})}) \right)$$

## Finding Topics

- We are given the set of words  $\mathbf{W}$  and don't really care about  $\tau_i^{(d)}$  the topic associated with word  $i$  in document  $d$
- But we are interested in the words associated with each topic  $\phi^{(t)}$
- And the topics associated with each document  $\theta^{(d)}$
- To compute them we need to sample the probability distribution
- One way to do this is using Monte Carlo methods (see next lecture)

## Summary

- Building probabilistic models is an intricate process
- Graphical models provide a representation showing the causal relationship between random variables
- This allows us to break down the joint probability of all the variables into conditional probabilities
- This is useful for building the model, but also can speed up evaluating expectations
- Making observations changes the probabilities of random variables
- It is possible to generate very rich models such as Latent Dirichlet Allocation (LDA)